
CSE4354 Real-time Operating Systems

Fall 2026

As the instructor for this course, I reserve the right to adjust this schedule in any way that serves the educational needs of the students enrolled in this course.

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Instructor Information

Instructor Name

Jason Losh, Ph.D.

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ERB 649

[View Campus Map](#)

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Email Address

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Faculty Profile

<https://mentis.uta.edu/explore/profile/jason-losh>

Office Hours

Office hours will be held before and after each class outside the classroom, since this time is most convenient to students. Office hours are also available by appointment.

Communication Guidelines

Communication with me outside class should be through e-mail. I will respond to emails on M-Th within 24 hours in most cases.

All class-wide communication by the instructor, including distribution of homework sets, will occur via the class listserv. If you are enrolled prior to the first day of class, you will be added to the listserv automatically. If you add on or after the first day of class, please sign up for the CSE4354-L listserv by sending an e-mail from your UTA e-mail account to listserv@listserv.uta.edu from your UTA e-mail account (no subject line needed) and the command SUBSCRIBE CSE4354-L as the message body. You will then receive an e-mail from the listserv server to which you must acknowledge to join the listserv with "OK" in an e-mail.

Course Information

Section Information

CSE 4354-001

Course Description

Implementation of a real-time operating system with cooperative and preemption context switching, priority scheduling, semaphores, message queues, and inter-process communications on bare metal microcontrollers. Course includes multiple projects with hardware construction and rigorous software requirements..

Prerequisites

C or better in both CSE 3320 and CSE 3442 or 5400.

Time and Place of Class Meetings

TTh 7:00-8:20pm, ERB 129

This is a 100% face-to-face course. This is not an online course.

Classroom/Lecture Recording Policy

Faculty maintain the academic right to determine whether students are permitted to record classroom and online lectures. Recordings of classroom lectures, if permitted by the instructor or pursuant to an ADA accommodation, may only be used for academic purposes related to the specific course. They may not be used for commercial purposes or shared with non-course participants except in connection with a legal proceeding.

Recording of classroom and online lectures in this course is not allowed.

Time Zone

This course operates on Central Time. All times listed for class meeting times, exams, and assignment deadlines are in Central Time (CT).

Student Learning Outcomes

Upon successful completion of this course, students will have knowledge of:

1. Real-time operation system design
2. ARM M4F Architecture, NVIC, and MPU operation
3. ARM 7 interrupts
4. Service calls for OS services
5. Task switching using non-deterministic pending service call
6. Thread safety, critical sections and atomic operations
7. Synchronization primitives, fences, and barriers
8. Inter-thread/process communications
9. Cooperative and preemptive OS design
10. Priority scheduling, priority inversion, and priority inheritance
11. Analysis of RTOS performance (deterministic limits, switching times)
12. Security and privilege

Textbooks and Other Course Materials

No textbook will be required for this course.

Extensive references, datasheets, application notes, and class notes will be provided on the course web site at <http://ranger.uta.edu/~jlosh/>.

Technology & Equipment Requirements

The computer and OS must be capable of running appropriate compiler tools for programming the microcontroller. Ubuntu is used in the labs and lectures.

Current computer recommendations are available at <https://www.uta.edu/academics/schools-colleges/engineering/students/student-computer>.

All students are required to have a TM4C123GXL evaluation kit. Information on this board is available at <http://www.ti.com/tool/EK-TM4C123GXL>. You can order from TI directly, mouser.com, or other parts distributors. The cost is around \$20 from TI.

Assignments and Examinations

Shell Nano-project: Tuesday, September 1
MPU Mini-project: Tuesday, October 6
Final Project: Tuesday, December 1

Expectations for Out-of-Class Study

For every credit hour earned, a student should spend 3 hours per week working outside of class. Multiply the number of credit hours for the course by 3 to determine study hours. A 3-credit hour course would require 9 additional study hours per week.]

Beyond the time required to attend each class meeting, students enrolled in this 3 credit-hour course should expect to spend at least an additional 9 hours per week of their own time in course-related activities, including reading required materials, completing assignments, etc.

Grading Information

Assignment Name	SLO #	Value (pts or %)
Nano Project*	1-2	10 pts
Micro Project*	1-3, 12	40 pts
Project*	1-12	50 pts
Total		100 pts

* If sufficient TA support is not possible, then the instructor reserves the right to change the project deliverables, including the possibility of one single project submission at the end of the semester.

Students are expected to track their performance throughout the semester, which Canvas facilitates, and seek guidance from available sources, including the instructor, if their performance drops below satisfactory levels. Refer to the [Student Support Services](#) section below.

Final Grade Calculation

Range (pts or %)	Letter Grade
90-100	A
80-89	B
70-79	C

Range (pts or %)	Letter Grade
60-69	D
0-59	F

Make-Up Exams & Late Work Policy

No late work is accepted.

Extra Credit Policy

Extra credit may be available as part of the labs.

Grades & Feedback Timeline

- The projects will consist of hardware construction and firmware development and it is expected that it will take approximately 90 hours to complete in total.
- All 3 projects are individual assignments. Discussing project topics is allowed, but the submissions must be unique. Sharing of code is not allowed.
- An interim deadline for hardware construction will apply.
- While the lecture content of this course is shared across multiple courses, students enrolled in a graduate course will have additional project components assigned.

Grade Grievances

Any appeal of a grade in this course must follow the procedures and deadlines for grade-related grievances as published in the current [University Catalog: Grades and Grading Policies](#).

University & Course Policies

UTA students are encouraged to review the below institutional policies and informational sections and reach out to the specific office with any questions. To view this institutional information, please visit the [Institutional Information](#) page (<https://resources.uta.edu/provost/course-related-info/institutional-policies.php>) which includes the following policies among others:

- Drop Policy
- Disability Accommodations
- Title IX Policy
- Academic Integrity
- Student Feedback Survey
- Final Exam Schedule

Attendance

Students should review the University Class Attendance Policies on the [Class Attendance Policies page](#). The following attendance policy will be applied in this course.

Generative AI Use in This Course

The use of Generative AI (GenAI) in course assignments and assessments must align with the guidelines established by the instructor. Unauthorized use of GenAI could result in breaches of academic integrity. Instructors are responsible for clearly delineating the permissible uses of

GenAI in their courses, underscoring the importance of responsible and ethical application of these tools.

[Community Standards](#) within the [Office of the Dean of Students](#) articulate the university's stance on [academic integrity and scholastic dishonesty](#). These standards extend to the use of GenAI. Unauthorized or unapproved use of GenAI in academic work falls within the scope of these policies and will be subject to the same disciplinary procedures.

As the instructor for this course, I have adopted the following policy on student use of GenAI. 

Approach	Description
Prohibition of GenAI Use	In this course, the focus is on the development of independent critical thinking and the mastery of subject-specific content. To ensure that all submitted work accurately reflects personal understanding and original thought, the use of Generative AI (GenAI) tools in completing assignments or assessments is strictly prohibited. This policy supports our commitment to academic integrity and the direct measurement of each student's learning against the course's Student Learning Outcomes (SLOs). Any work found to be generated by AI will be subject to academic review.

Institutional Policies

UTA students should review the [University Catalog](#) and the [Syllabus Institutional Policies](#) page for institutional policies and contact the specific office with any questions. The institutional information includes the following policies, among others:

- Drop Policy
- Disability Accommodations
- Academic Integrity
- Electronic Communication

UTA Honor Code

UTA students are expected to adhere to and observe standards of conduct compatible with the University's functions as an educational institution and live by the [University of Texas at Arlington's Honor Code](#). It is the policy of The University of Texas at Arlington to uphold and support standards of personal honesty and integrity for all students consistent with the goals of a community of scholars and students seeking knowledge and responsibility.

Student Support Services

Student Services Page

The [Student Services page](#) provides links to many resources available to UTA students, including:

Academic Success

Counseling and Psychological Services (CAPS)
Health Services
Students with Disabilities
Veteran Services

Students are also encouraged to check out [Career Center](#) resources to enhance their career-readiness, find student employment, search for internships, and more. We encourage [Major Exploration](#) and the use of [Experiential Major Maps](#) to keep students on track for graduation. Refer to the [Graduation Help Desk](#) for more details.

Accessibility of Course Materials

Some course materials, such as PDFs of musical scores, technical drawings, graphs, blueprints, design plans, or artworks (common in fields like drawing, painting, or construction drafting), may not fully comply with all [Web Content Accessibility Guidelines \(WCAG\)](#) requirements.

The University of Texas at Arlington is dedicated to ensuring all students have equal access to information. If you experience any accessibility barriers with course materials, please know that accommodations are available. You can get assistance through the [Student Access and Resource \(SAR\)](#) Center or by contacting your instructor directly. Please don't hesitate to reach out if you need help.

Online Academic Success Guide

Visit the [Online Academic Success Guide](#) to explore a list of helpful tips and resources to help you succeed in your online journey.

UTA Health and Wellbeing Resources

UT Arlington is committed to the safety, success, and well-being of our students. To support our community, UTA has established a Community Advocacy, Response, and Engagement (CARE) Team, a dedicated group of campus professionals responsible for helping students who could benefit from academic, emotional, or psychological support, as well as those presenting risks to the health or safety of the community. If you know of someone experiencing challenges, appearing distressed, needing resources, or causing a significant disruption to the UTA community, please submit a [CARE Referral](#) by visiting the [CARE Team](#) page. You may also submit a referral for yourself if you would like additional support.

UTA students also have access to virtual, on-demand emotional support, appointment-based counseling, advanced psychiatric care, and more. For more information, visit [TimelyCare](#).

NOTE: If a person's behavior poses an immediate threat to you or someone else, contact UTA Police at 817-272-3003 or dial 911. If you or someone you know needs to speak with a crisis counselor, please reach out to the [MAVS TALK 24-hour Crisis Line](#) at 817-272-8255 or the [National Suicide and Crisis Lifeline](#) at 988.

Librarian to Contact

Each academic unit has access to [Librarians by Academic Subject](#) who can assist students with research projects, tutorials on plagiarism, citation references, as well as support with databases and course reserves.

Course Schedule

Class Date(s)	Topic(s)	Materials	Assignments Due
Week 1-3	Nano Project	Read Project materials and relevant section of the datasheets covered in class. Read and understand the supplied code developed in class.	Day 6
Week 8	Micro Project	MPU, NVIC, and related OS topics	Day 15
Weeks 9-16	Project	Read project materials and relevant section of the datasheets covered in class. Read and understand the supplied code developed in class.	Day 29

Safety Information & Resources

Lab Safety Safety Rules for ERB 121-127 and 132 Labs:

Scope:

- All UTA safety rules and regulations must be followed.
- These rules are in addition to UTA lab safety rules.
- In the event that a rule contained below is in conflict with UTA lab safety rules, the UTA safety rules shall supersede.

General rules:

- Students can only be in the lab when a teaching assistant (TA), faculty member, or staff member is present.
- Students should be professional at all times in the lab.
- Food is not allowed in the lab at any time. Drinks, including those in sealed container, is not allowed in the lab. The only exception to this rule is the marked tables in ERB 125.
- When leaving the lab, all work surfaces and floors should be clear of breadboards, cables, wires, and tools prior to leaving.
- When leaving the bench, turn off all lab equipment and unplug soldering irons.

- When leaving the bench, make sure that all cables, tools, and soldering equipment are properly stored in the correct location.
- Please ensure that the lab is kept in a neat and tidy manner.
- Please pick up any loose wires or parts on the bench and floor before leaving the lab.
- Note any hazards observed in the lab to the TA, faculty, or staff member immediately.
- For test equipment incorporating multi-language menus, such as oscilloscopes, please return the language to English before leaving the bench.
- When returning parts that are not consumables, make certain that the parts are returned to the correct drawer. If you are not certain, please leave them with the TA, faculty member, or staff member.
- Students should store backpacks and similar items in a way that does not create a trip hazard to others. In ERB 126 and 127, there are cabinet spaces at the base of the benches for this purpose.

Soldering irons:

- Soldering must be performed in the labs only at the soldering benches. Never solder in dorm rooms.
- Soldering irons should be used with care, while wearing safety glasses, and only after receiving training.
- When soldering and removing parts, or reworking a board, please use special care to ensure that solder is not splattered.
- Soldering irons must be placed back in their soldering station holder when not soldering to prevent the chance of injury or fire.
- Please keep the soldering station sponges wet when cleaning the iron tip but ensure that water is not spilled on the floor creating a slip hazard.
- Use the soldering iron smoke absorber fan units when soldering. Use them in the horizontal position (air exits upward) to prevent directing air flow across the table into the face of another user.
- Some solders can contain lead, so wash hands thoroughly after using the soldering irons. No eating or drinking is allowed at the soldering benches. The green solder spools in the lab generally indicate a lead-free solder.
- No self-contained butane soldering irons are permitted.
- Wear appropriate personal protection equipment (PPE).

Chemicals and lasers:

- In labs where chemicals or lasers are used, students and faculty must receive the appropriate safety training prior to working in the lab.
- In labs with chemicals, consult the safety data sheet (SDS) folder for information.
- Wear appropriate personal protection equipment (PPE) at all times.
- If transferring chemicals to secondary containers, clearly mark the contents of the container. Specially, a label with spelled out chemical name and hazards is required. Also, for water bottles, label as “not for human consumption on the bottle.” A green dot can also be attached to indicate that the chemical is not hazardous.

Hand tools:

- Hand tools must be used with care and only when safety glasses are being worn.
- Diagonal cutters in particular can create tension on the wires during the cutting process, ejecting the loose wire, so please use special care.
- Wear appropriate personal protection equipment (PPE).

Power tools:

- Short time use of small powered cutting tools such as a drill/driver can only be used at the soldering tables using a backup board to prevent damage to the tables.
- For extended machining tasks, please use the designated Makerspace areas that are designed to handle the additional safety requirements and dust inhalation hazards.
- Wear appropriate personal protection equipment (PPE).
- Jewelry, necklaces, and lanyards should be removed.
- Long hair should be tied back to prevent being caught in the tool.
- For labs with drill presses, band saws, laser cutters, CNC machines, and similar equipment, students must take the appropriate safety training prior to using the equipment.

Electrical hazards:

- The labs for these classes generally use voltages of 32V or less, but care must always be shown in using electrical circuits, regardless of the voltage.
- Do not use voltages of more than 32V unless approved in writing by the instructor.
- Do not modify the wiring or attempt repair of any lab equipment.
- Most of the lab equipment operates from 120V AC, which is a lethal voltage. Never pull on a cord to unplug it as this can cause damage to the strain relief and insulation, potentially resulting in exposed electrical conductors.
- Please notify the TA, faculty member, or staff member and stop using the equipment immediately if you see nicks or damage to a power cord or other hazardous conditions.

Computers:

- Students should not install any software on the lab computers without approval of the TA, faculty member, or staff member.
- Students should not remove any of the cables on the computer and the monitor on the bench.
- In some labs, an HDMI cable is wired into each workstation for configuring Raspberry Pi and similar computer hardware. This cable should not be disconnected from the monitor.

No device connected to the wired network of a lab should also have a WiFi connection enabled as this represents a security risk. This includes both personal student laptops and activating WiFi direct on printers.
